

Case Study II: Styrian Health Data

Week 5 Update: Spatial analysis and Model Improvements

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Study Overview & Objectives

Study Focus:

- ▶ This study applies spatial analysis to patient-level BP data across Austria to uncover geographic patterns.

Research Questions:

1. Does **blood pressure vary** by **urban-rural typology**?
2. Is there **spatial clustering of hypertension** across districts?
3. Does **regional GDP (NUTS3)** correlate with **blood pressure levels**?

Data Sources

- ▶ Patient-level **BP records** linked to Austrian administrative geography (**Bezirk/NUTS3**)
- ▶ Open-source regional indicators from <https://www.statistik.at/atlas/>
 - ▶ Gross Domestic Product per Capita - NUTS3
 - ▶ Divisions according to urban and rural areas

Data Limitations

- ▶ **Urban–Rural match rate:** Municipality-level matching performed using the Eurostat urban–rural typology CSV.
- ▶ Some *Gemeinden* could not be matched directly due to post-2015 Austrian municipal mergers (e.g., Bruck an der Mur + Mürzzuschlag → Bruck-Mürzzuschlag; Hartberg + Fürstenfeld → Hartberg-Fürstenfeld).
- ▶ **Patient threshold:** Only districts and NUTS3 regions with ≥ 30 patients were retained for spatial and GDP-related analyses.
- ▶ Grey districts/regions on maps indicate missing data: either no patients recorded in that area or unresolved municipality matching.

Module 1 — Urban-Rural Typology

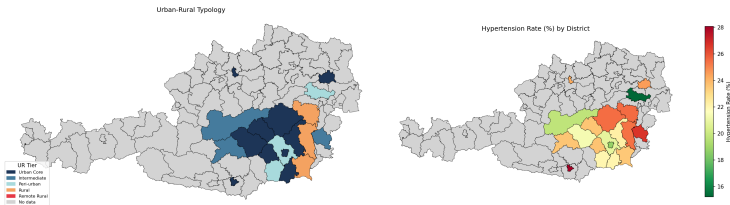
- ▶ Patients matched to Eurostat urban-rural typology (*Gemeinde* level)
- ▶ Five tiers: Urban Core, Intermediate, Peri-urban, Rural, Remote Rural
- ▶ District classification based on modal urban-rural tier
- ▶ Hypertension:

$$\text{SBP} \geq 140 \text{ mmHg}$$

- ▶ Spatial weights: Queen contiguity (row-standardised)
- ▶ Spatial tests: Moran's I and LISA

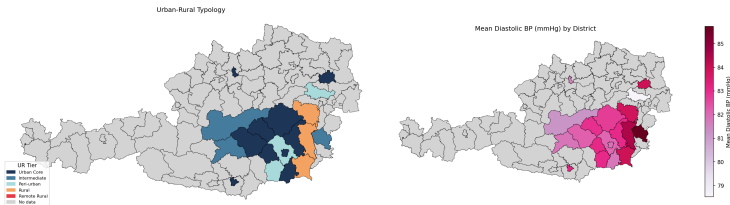
Urban-Rural Typology vs Hypertension

Urban-Rural Typology vs Hypertension Rate (%) — Austria (District Level)



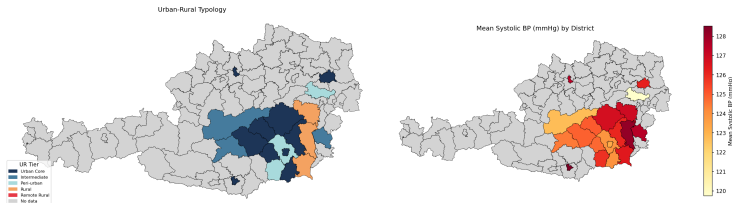
Urban-Rural Typology vs Mean Diastolic BP

Urban-Rural Typology vs Mean Diastolic BP (mmHg) — Austria (District Level)



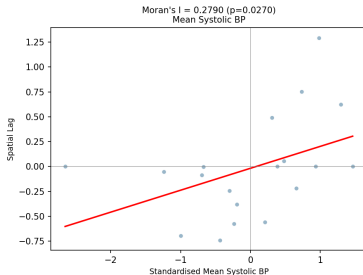
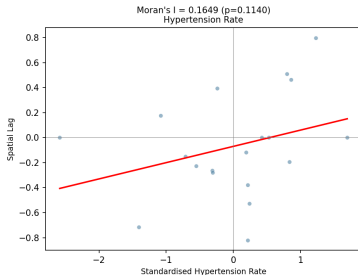
Urban-Rural Typology vs Mean Systolic BP

Urban-Rural Typology vs Mean Systolic BP (mmHg) — Austria (District Level)



Spatial Autocorrelation Analysis (Moran's I)

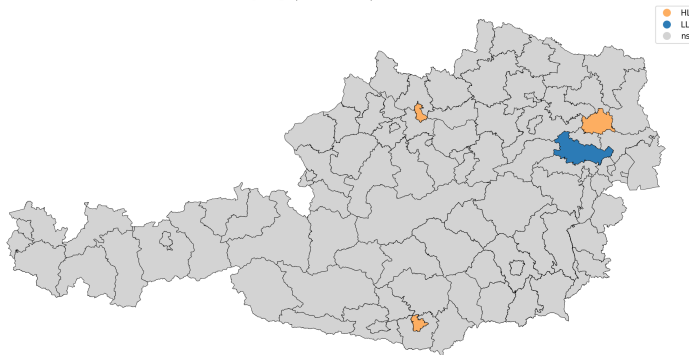
Global Spatial Autocorrelation — Moran's I (District Level)



- ▶ Urban–rural typology alone does not explain the spatial pattern.
- ▶ Higher systolic BP, diastolic BP, and hypertension prevalence are concentrated in eastern and southeastern Austrian districts.
- ▶ Elevated rates occurred across urban, intermediate, and rural districts
- ▶ Neighboring districts tend to show similar cardiovascular risk profiles, indicating spatial clustering.
- ▶ Moran's I confirms significant positive spatial autocorrelation for hypertension prevalence ($I = 0.154$, $p = 0.025$).

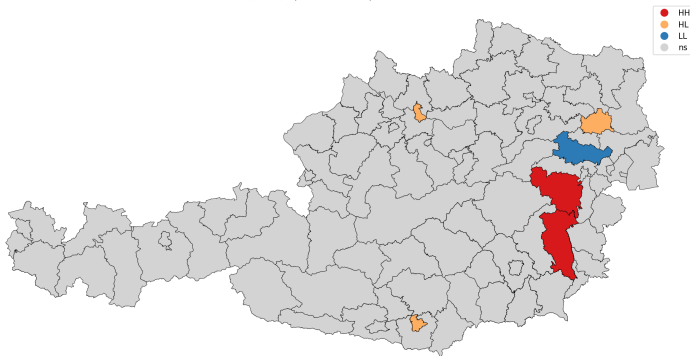
LISA Cluster Findings (District Level)

LISA Cluster Map — Hypertension Rate (District Level)
HH=High-High | LL=Low-Low | HL/LH=Outliers



LISA Cluster Findings (District Level)

LISA Cluster Map — Systolic BP (District Level)
HH=High-High | LL=Low-Low | HL/LH=Outliers



LISA Cluster Summary — Key Spatial Findings

Hypertension Rate

- ▶ **LL Cluster — Baden**
 - ▶ Low hypertension surrounded by low neighbouring districts
 - ▶ Stable low-risk zone
- ▶ **HL Urban Outliers**
 - ▶ Wien, Linz, Klagenfurt
 - ▶ Higher hypertension than surrounding districts

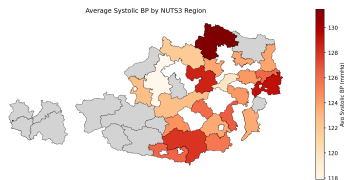
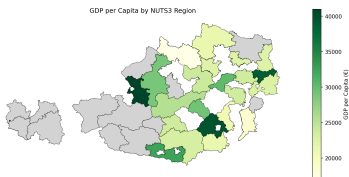
Systolic BP

- ▶ **HH Cluster**
 - ▶ Hartberg-Fürstenfeld & Neunkirchen
 - ▶ Significant southeastern high-BP cluster
- ▶ **LL Cluster — Baden**
 - ▶ Consistent low-BP spatial zone
- ▶ **HL Urban Outliers**
 - ▶ Wien, Linz, Klagenfurt

Spatial patterns suggest both regional clustering and urban outlier effects in cardiovascular risk.

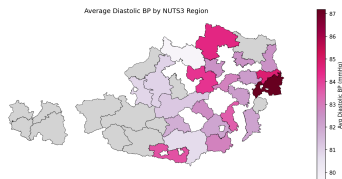
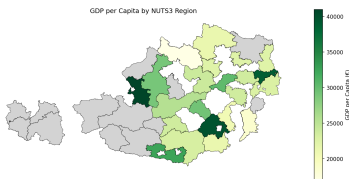
Regional GDP per Capita (NUTS3) vs BP

Geographic Distribution: Wealth vs Systolic BP (NUTS3)



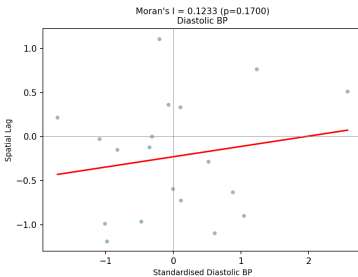
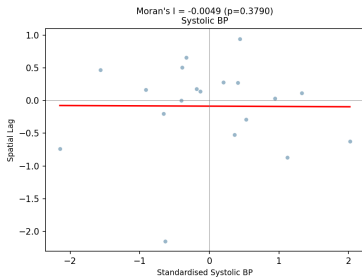
Regional GDP per Capita (NUTS3) vs BP

Geographic Distribution: Wealth vs Diastolic BP (NUTS3)



Spatial Autocorrelation of Blood Pressure (NUTS3 Level) - GDP

Global Spatial Autocorrelation — Moran's I (NUTS3 Level)



GDP and Blood Pressure — Key Findings

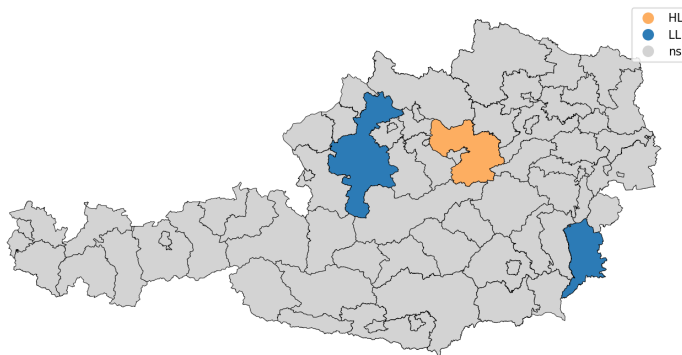
- ▶ Western and central Austrian regions generally showed higher GDP levels with comparatively lower or moderate blood pressure values.
- ▶ Eastern and southeastern regions exhibited higher systolic BP and hypertension prevalence despite mixed economic profiles.
- ▶ Some economically strong urban regions such as Wien and Linz-Wels also showed elevated BP levels, suggesting that higher GDP alone does not protect against cardiovascular risk.
- ▶ Correlation analysis showed only a weak relationship between GDP and systolic BP:

$$r = -0.064, \quad p = 0.784$$

- ▶ Spatial autocorrelation at the NUTS3 level was not statistically significant, indicating no strong regional clustering pattern linked to GDP.

Local Spatial Clusters of Systolic BP (NUTS3 Level)

LISA Cluster Map — Systolic BP (NUTS3 Level)
HH=High-High | LL=Low-Low | HL/LH=Outliers



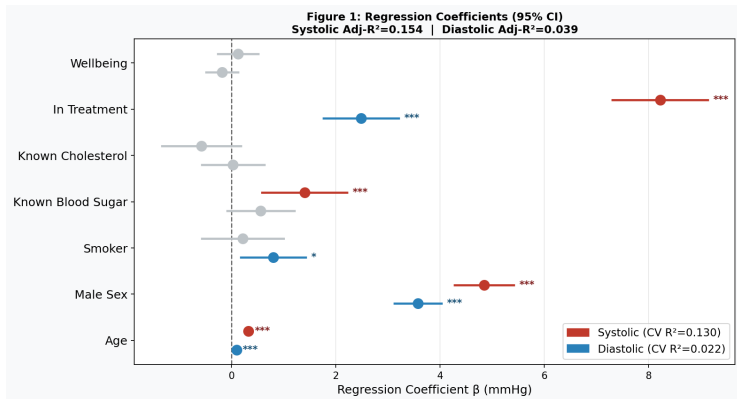
NUTS3 Spatial Clustering — Key Findings

- ▶ Although global spatial autocorrelation is not significant at the NUTS3 level ($I_{\text{sys}} = -0.005$, $I_{\text{dia}} = 0.123$), local clustering is still observed.
- ▶ A few regions in central-western and southeastern Austria show Low-Low (LL) BP clusters.
- ▶ Some central and northeastern regions appear as High-Low (HL) outliers, indicating elevated BP relative to neighbouring regions.
- ▶ These findings suggest that BP spatial structure becomes weaker at broader regional aggregation levels compared with district-level analysis.

Classical Modelling Approach

Method

- ▶ Ordinary Least Squares (OLS) linear regression.
- ▶ Separate models fitted for:
 - ▶ Mean systolic BP
 - ▶ Mean diastolic BP
- ▶ Predictors evaluated using:
 - ▶ Regression coefficient (β)
 - ▶ p-value (statistical significance)
 - ▶ 95% confidence interval
- ▶ Model performance assessed using 5-fold cross-validation (R^2)

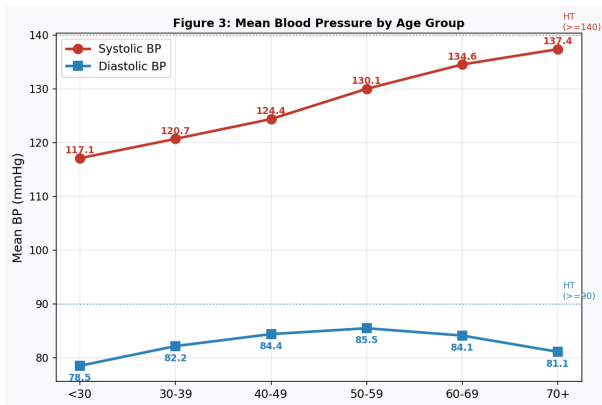


- Age, Male Sex, and In Treatment are the strongest predictors — all significant at $p < 0.001$. Cholesterol, Wellbeing, and Smoker showed no significant effect on systolic BP.

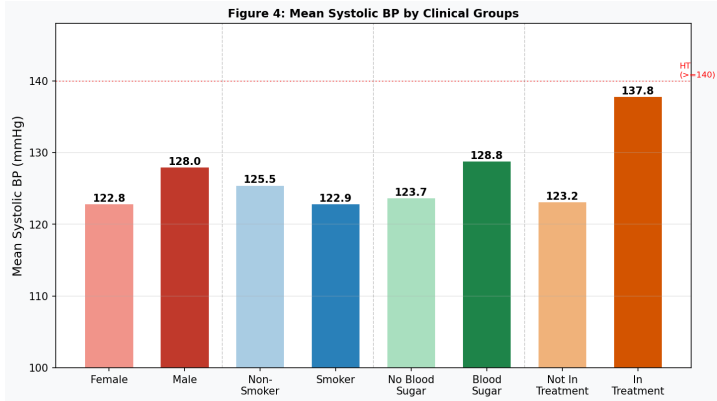
Figure 2: Coefficient Summary
(β = mmHg per unit | *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$)

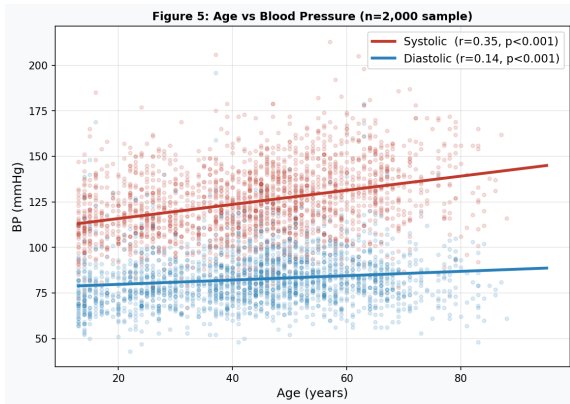
Predictor	β Systolic	p-value	β Diastolic	p-value
Age	+0.321	<0.001***	+0.098	<0.001***
Male Sex	+4.846	<0.001***	+3.578	<0.001***
Smoker	+0.211	0.599 ns	+0.801	0.011*
Known Blood Sugar	+1.400	<0.001***	+0.560	0.089 ns
Known Cholesterol	-0.583	0.131 ns	+0.027	0.929 ns
In Treatment	+8.221	<0.001***	+2.483	<0.001***
Wellbeing	+0.125	0.528 ns	-0.180	0.249 ns

- Being in treatment is associated with +8.2 mmHg higher systolic BP. Male sex adds +4.8 mmHg. Every one year of age adds +0.32 mmHg.



- ▶ systolic BP rises from 117 mmHg in people under 30 to 137 mmHg in people over 70. Diastolic BP peaks around 50 to 59 then slightly drops in older age.





Linear Regression — Summary

- ▶ **Age:** Each additional year increases systolic BP by +0.32 mmHg and diastolic BP by +0.10 mmHg ($p < 0.001$).
- ▶ **Sex:** Males have higher BP than females — +4.8 mmHg systolic, +3.6 mmHg diastolic ($p < 0.001$).
- ▶ **Treatment Status:** Patients currently in treatment show highest BP — +8.2 mmHg systolic, +2.5 mmHg diastolic (strongest predictor; likely reverse causation).
- ▶ **Known Blood Sugar:** Associated with higher systolic BP (+1.4 mmHg); diastolic not significant after adjustment.
- ▶ **Smoking:** Increases diastolic BP (+0.80 mmHg, $p = 0.011$); no significant effect on systolic BP.
- ▶ **Cholesterol & Wellbeing:** No significant effect on either BP measure after controlling for other predictors.